Mohammed Saifan

SOLUTIONS ARCHITECT & SYSTEMS ENGINEER

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SUMMARY

15-year-old systems builder and full-stack engineer from Bangalore with proven track record across hackathon wins, live production clients, and drone hardware systems. Specializes in reactive distributed architectures, high-performance WebGL compute, and hardware-layer ESP32 microcontrollers.

KEY PROJECTS & SYSTEMS ARCHITECTURE

AirGuard — Autonomous Pre-Flight UAV Avionics — State Runner-Up (2026)

TechnoFest Hackathon 2026 (2nd Place out of 100+ Karnataka teams)

- Engineered a pre-flight flight-safety layer with real-time geospatial dynamic geofencing and ADS-B meteorological radar.
- Integrated Google Gemini LLM API for automated FAA/DGCA aviation compliance validation in real time.
- Stack: React, TypeScript, Leaflet, Convex, Gemini API.

Strategos — Game Theory & Evolutionary Dynamics 3D Engine — Simulation Lab

WebGL Mathematical Engine

- Architected real-time 3D simulation exploring multi-agent Prisoner's Dilemma, Nash Equilibria, and cellular mutation vectors.
- Maintains continuous 60 FPS WebGL simulation runs completely client-side in the browser with zero backend latency.
- Stack: React 19, TypeScript, Three.js, Vitest, WebGL.

MBSYS & Hunar — Enterprise Platforms — Production Client Work

Live Business Clients in Bangalore (Serving 2 Enterprises)

- Architected enterprise IT infrastructure web platform with real-time CMS, GPU-accelerated canvas, and synchronized multi-brand transitions.
- Achieved 99.9% uptime with sub-50ms reactive response times across Bangalore corporate users.
- Stack: Next.js, React 19, Convex, Tailwind CSS.

ESP32 Hardware Flight Avionics & Drone Telemetry — Embedded Systems

Silicon-to-UI Architecture

- Authored custom C++/FreeRTOS firmware on ESP32 microcontrollers for MPU6050 6-axis IMU attitude stabilization.
- Implemented low-latency WebSocket telemetry transmission connecting flight hardware directly to web dashboards.

HONORS & AWARDS

- **State 2nd Place / Runner-Up** — TechnoFest Hackathon 2026 (AirGuard autonomous UAV safety system).
- **National Finalist** — AI for Bharat Hackathon 2026 (BBIE Forensic Enterprise Registry).
- **Production Client Deliveries** — 2 live commercial client platforms deployed in Bangalore.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, C++, Python, HTML5, Modern CSS, SQL
Frontend & 3D:	React 19, Next.js (App Router), Three.js, WebGL Shaders, Tailwind CSS, Framer Motion
Backend & Cloud:	Convex (Reactive State), Supabase, Node.js, Netlify Serverless, WebSockets, REST
Hardware Layer:	ESP32, FreeRTOS, MPU6050 IMU, Arduino C++, UAV Avionics, I2C / SPI Sensors